

5.4 Impacts of Computing in Various Fields

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 09 of 12

5.4 Impacts of Computing in Various Fields

LEARN IT

Computing has drastically changed society. It makes life **faster and more connected**, but it also brings challenges like **privacy risks** and **technology addiction**. For every field, think of both sides.

Field	Positive impact (+)	Negative impact (–)
Education	Online classes, e-books and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain).
Communication	Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai).	People interact less in person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Smart and responsible use

- **Protect personal information** and use **strong passwords** (long, with letters, numbers and symbols; never share your OTP or CNIC details with callers).

- **Do not share fake news** or unverified WhatsApp forwards.
- **Balance screen time** with physical sports and activities.

WORKED EXAMPLE

Scenario analysis — a bank in Karachi moves to mobile banking:

Step 1 — Identify the field: Business (banking).

Step 2 — Positive: Customers transfer money in seconds from home; no queue at the branch; the bank processes more transactions with fewer staff (speed and efficiency).

Step 3 — Negative: Some cashier jobs are no longer needed (job loss); hackers may attempt cyber-attacks or phishing SMS to steal accounts.

Step 4 — Responsible use: Customers should use strong passwords, never share OTPs, and report suspicious messages.

REMEMBER IT

For the five fields, remember "**E-C-B-H-A**" — **E**ducation, **C**ommunication, **B**usiness, **H**ealthcare, **A**griculture — and always write one + and one – for each.

COMMON MISTAKE

Mistake: Writing only positive impacts in a "discuss the impacts" question. **Fix:** "Impact" means *both* sides. Examiners expect a balanced answer: advantages, disadvantages, and how to use technology responsibly.

اردو میں سمجھیں

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔ تعلیم، مواصلات، کاروبار، صحت اور زراعت میں اس نے کام تیز اور آسان بنائے ہیں، لیکن اس کے ساتھ اسکرین کی لت، سماجی تنہائی، ملازمتوں کا خاتمہ، ہیکنگ اور پرائیویسی کے خطرات بھی پیدا ہوئے ہیں۔ ذمہ دارانہ استعمال کے لیے مضبوط پاس ورڈ رکھیں، جعلی خبریں نہ پھیلائیں، اور کھیل کود کے ساتھ اسکرین ٹائم میں توازن رکھیں۔

CHECK YOURSELF

1. Give one positive and one negative impact of computing in healthcare.
2. Why can't many farmers in Pakistan benefit from smart agriculture tools?
3. State three rules of smart and responsible use.

Answers: 1. + Digital records and telemedicine help treat remote patients; – patient data privacy is at risk if servers are hacked. 2. Poor internet access in rural areas. 3. Protect personal information/use strong passwords; do not share fake news; balance screen time with physical activity.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Computing in health:

- a. Records, imaging, telemedicine
- b. Only games
- c. Only NAND
- d. Only sprints

MCQ 2. Computing in education:

- a. LMS, simulations, analytics
- b. Only chalk forever
- c. Only OSI layer 1
- d. Only keys

MCQ 3. A social harm is:

- a. Addiction / fake news / job loss
- b. Faster search in a library
- c. Safer banks always only
- d. Shorter SDLC only

MCQ 4. E-waste is:

- a. Discarded electronics
- b. A Python error
- c. A foreign key
- d. A histogram bin

MCQ 5. The exam pattern is:

- a. One field + one good + one bad + one control
- b. Only definitions
- c. Only code
- d. Only careers

MCQ 6. Banks use computing for:

- a. Accounts, ATMs, fraud analytics
- b. Growing wheat
- c. Drawing K-maps
- d. Sorting hats

Extra short questions

1. Two fields and one impact each.
2. One environmental issue.
3. One social issue.
4. One control/mitigation.

Extra long question

1. "Computing has only benefits." Refute with three fields, each with a benefit, a harm, and a mitigation. 10 marks.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Health: telemedicine. Transport: ride apps / signals. Banking: mobile wallets.
2. E-waste, energy of data centres.
3. Digital divide — students without smartphones.
4. Laws, privacy settings, recycling, lab access at college.

Long

1. Health, education, media. Structure a table. Close: technology is a tool; policy and literacy decide the outcome.

Next lecture: 5.5 Assistive Technologies